

4 Breathing out as hard as you can is called forced expiration.

0 4 . 1 Describe and explain the mechanism that causes forced expiration.

[4 marks]

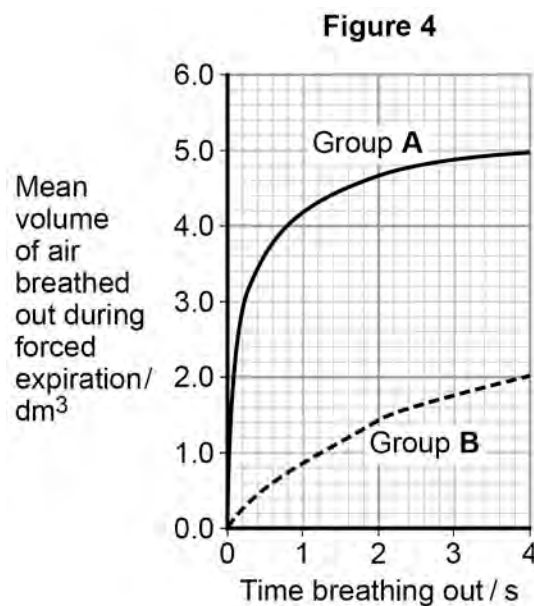
Two groups of people volunteered to take part in an experiment.

- People in group **A** were healthy.
- People in group **B** were recovering from an asthma attack.

Each person breathed in as deeply as they could. They then breathed out by forced expiration.

A scientist measured the volume of air breathed out during forced expiration by each person.

Figure 4 shows the results.



- 0 4** . **2** Forced expiration volume (FEV) is the volume of air a person can breathe out in 1 second.

Using data from the first second of forced expiration, calculate the percentage decrease in the FEV for group **B** compared with group **A**.

[1 mark]

Answer = _____ %

- 0 4** . **3** The people in group **B** were recovering from an asthma attack. Explain how an asthma attack caused the drop in the mean FEV shown in **Figure 4**.

[4 marks]

[Extra space] _____
